HR Analytics Dashboard

1. Background

This project analyzes employee attrition data for a service-based company. The company has around 10,000 employees and is facing employee turnover challenges that may affect project delivery, workforce stability, and productivity.

The goal of this dashboard is to identify key factors influencing employee attrition and support better HR decision-making through data visualization.

2. Business Problem

The main business problem is to understand why employees are leaving the organization and which factors are contributing most to attrition.

Key questions explored in this project:

1. What is the overall attrition rate?
2. Which departments have the highest attrition?
3. How does job role influence attrition?
4. Is overtime contributing to higher attrition?
5. Does business travel affect employee attrition?
6. How do job satisfaction and work-life balance affect retention?
7. Is distance from home related to employee attrition?
8. Are salary hikes helping improve employee retention?
9. How does education level relate to attrition?
10. Which job levels and job roles require more HR attention?

3. Tools Used

- Power BI
- SQL Server
- DAX
- Excel / CSV Dataset
- Data Visualization

4. Dataset Overview

The dataset includes employee-level HR information such as employee ID, age, department, job role, job level, salary, attrition status, overtime, business travel, job satisfaction, work-life balance, environment satisfaction, years of experience, and years since last promotion.

5. Dashboard Overview

The Power BI dashboard includes key HR metrics such as:

- Total Employees
- Total Attrition
- Active Employees
- Attrition Rate
- Average Age
- Attrition by Department
- Attrition by Job Role
- Attrition by Overtime
- Attrition by Business Travel
- Attrition by Salary Hike
- Attrition by Education
- Attrition by Distance from Home
- Job Satisfaction, Work-Life Balance, and Work Environment analysis

6. Solution Approach

First, the employee data was loaded and cleaned. SQL was used to create the employee table and insert records. After that, the dataset was imported into Power BI for data modeling and visualization.

DAX measures were created to calculate important KPIs such as total employees, total attrition, active employees, attrition rate, and average age. Different charts were then used to analyze attrition trends across departments, job roles, overtime, business travel, salary hike, education, and employee satisfaction factors.

7. Key Insights

The dashboard shows that employee attrition is influenced by multiple factors such as overtime, job satisfaction, work-life balance, business travel, salary hike, and department-level differences. These insights can help HR teams identify high-risk employee groups and improve retention strategies.

8. Conclusion

This HR Analytics Dashboard provides a clear overview of employee attrition patterns and helps the organization understand where employee turnover is high. The dashboard can support HR managers in making data-driven decisions to improve employee retention, workforce planning, and organizational stability.